Sahaj Bhadja

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Summary

Software engineer with 2+ years shipping production Python services, LLM workflows, REST APIs, webhooks, and customer-facing integrations. Experienced in translating requirements into reliable systems, owning delivery through testing, monitoring, debugging, and deployment.

Education

University of Southern California, Los Angeles, CA **August 2024 – May 2026**
Master of Science in Computer Science *GPA: 3.75/4*
Coursework: Analysis of Algorithms, Database Systems, Web Technologies, Advanced NLP

Pandit Deendayal Energy University, Gandhinagar, India **November 2020 – May 2024**
Bachelor of Technology in Computer Science and Engineering *GPA: 3.88/4*
Coursework: Software Engineering, Operating Systems, Object-Oriented Programming, Data Retrieval

Experience

USC Dornsife **Los Angeles, CA**
Software Engineer *September 2025 – Present*
- Reduced user-facing dashboard latency from 1.2 seconds to 350 milliseconds for 500+ users by profiling Python API request paths and tuning Azure SQL queries, indexes, and secure access patterns.
- Built FastAPI services and fault-tolerant asynchronous LLM workflows for 5,000+ monthly jobs, using explicit states, retries, reconciliation, and failure isolation to triple throughput while sustaining 99%+ reliable production processing.

Grad8 **Los Angeles, CA**
Software Engineer Intern *June 2025 – August 2025*
- Developed Flask APIs and React implementation workflows supporting 150+ users and 10,000+ daily requests, integrating secure ingestion, authentication, AWS RDS persistence, status tracking, and instructor-facing product experiences.
- Designed concurrent processing for 1,000+ uploaded files with real-time job-state updates, automated validation, and fault-tolerant execution, reducing turnaround 60% while maintaining stable API performance during peak demand.

USC AD Labs **Los Angeles, CA**
ML Research Assistant *January 2025 – May 2025*
- Built reproducible simulation and benchmarking pipelines processing 100,000+ records, applying structured automated tests, instrumentation, and systematic debugging to improve long-horizon planning accuracy 18% across controlled evaluations.
- Participated in cross-functional collaborative design reviews and validation experiments, documenting technical decisions and tuning a performance-critical architecture that improved closed-loop stability 15% across repeated controlled simulation scenarios.

Aasma Technology Solutions **India**
Software Developer Engineer – SecureSync *August 2023 – July 2024*
- Maintained 20+ automated unit and integration suites using Jenkins, PyTest, and Mockito across Windows, Linux, and macOS, accelerating defect diagnosis and improving confidence in cross-platform production releases.
- Partnered with Microsoft and GitHub stakeholders on API integrations, then optimized filtering, pagination, indexing, and caching to improve client response times 30% and production data-pipeline efficiency 68%.

LST Capitals **India**
Web Developer Intern *November 2022 – January 2023*
- Designed the end-to-end system workflow, mapped 12+ integration points, and presented the architecture to a client before building cross-platform booking experiences handling more than 50,000 records.
- Developed Flask APIs for bookings, pricing, and payments, delivering sub-200-millisecond responses and webhook-driven synchronization while improving confirmation speed 40% across customer-facing mobile and web experiences during peak demand.

Skills

Languages & Backend: Python, TypeScript, JavaScript, Java, C++, Flask, FastAPI, Node.js, REST APIs, JSON, object-oriented programming
Integrations & AI: webhooks, LLM orchestration, API integration, event-driven workflows, WebSockets, authentication, secure sessions, microservices, concurrency, asynchronous processing, real-time synchronization
Data & Cloud: AWS, Azure, Firebase, AWS RDS, PostgreSQL, MySQL, SQL Server, Azure SQL, S3, Blob Storage, indexing, caching, latency tuning
Delivery & Quality: Git, CI/CD, Jenkins, GitHub Actions, Docker, PyTest, Mockito, unit testing, integration testing, QA, observability, debugging, monitoring, documentation, Agile, production support, client collaboration

Projects

FinNexus | Live | *TypeScript, WebSockets, MongoDB, Kubernetes*
- Built a real-time financial platform supporting 50,000+ users and one million transactions, integrating secure third-party bank APIs, WebSocket synchronization, advanced filtering, and reliable scalable containerized production deployment.

HMIS | Live | *TypeScript, REST APIs, Appwrite, Twilio, Sentry*
- Built secure authentication and consent-based API workflows with Twilio notifications and Sentry production monitoring, improving reliability, debugging visibility, and responsiveness across customer-facing digital appointment and service experiences.

Big-Brain | Live | *Python, PostgreSQL, Clerk, GCP, REST APIs*
- Engineered secure document ingestion, user authentication, and cloud indexing workflows processing 50,000+ records, implementing reliable incremental updates and API performance tuning to reduce retrieval latency 40%.